# DPM — Deployment Planning Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

Architecture: SLGA™ — Software Lifecycle Governance Architecture

Parent Standard: SDPS — Software Deployment Standard

Category: Governance & Enforcement

Subcategory: Software Deployment Governance

Type: Core Module

Governed Space: Software Deployment Planning Governance

Module ID: SLGA-SDPS-DPM-01

Origin ID: OOF-SLGA-SDPS-DPM

Version: 1.0

Status: Canonical Open

Effective Date: 14.10.2026

Compatibility: Methodology OS — Modular Compatibility

GIE-Compatible: Yes

AI-Readable: Yes

Authority: OOF® — OriginOpen® Foundation

Protection: Protected Methodological Work

Canonical Language: English

## Canonical Definition

DPM — Deployment Planning Module is the Core Module within SDPS that governs how an identifiable software release is prepared for deployment into a defined operational environment through the establishment of a deployment purpose, target environment, deployment scope, execution strategy, operational prerequisites, resource requirements, verification conditions, intervention provisions, and rollback or recovery planning references.

## DPM establishes deployment planning as an identifiable, evidence-supported,

and reassessable governance mechanism rather than an informal technical preparation activity.

## It requires the intended deployment to remain distinguishable from its

authorization, execution, resulting production state, and operational acceptance.

## The module applies to manually planned deployments, automated delivery

pipelines, autonomous AI software agents, cloud-native infrastructure, distributed systems, embedded devices, and robotic platforms.

Its principal governance output is the Software Deployment Plan Profile.

## Operational Role

## DPM establishes the planning conditions required for a software release to

be considered for deployment into a particular operational environment.

It governs:

• Deployment purpose and intended outcome

• Release-to-deployment association

• Target environment identification

• Deployment scope and boundaries

• Deployment strategy selection

• Deployment sequence and dependencies

• Operational prerequisites

• Resource and capability requirements

• Execution actor and tool requirements

• Deployment timing and coordination

• Verification and acceptance conditions

• Intervention and escalation provisions

• Rollback and recovery planning

• Planning exceptions and uncertainties

• Deployment plan changes

• Planning evidence and traceability

• Plan readiness classification

• Handoff to deployment authorization

DPM does not authorize deployment, execute deployment actions, verify the resulting production state, or independently establish recovery success.

## Operational Question

How can an identifiable software release be prepared for deployment into a defined operational environment through a sufficiently specified, feasible, evidence-supported, and reassessable plan that preserves deployment purpose, scope, prerequisites, execution boundaries, verification conditions, and recovery provisions without confusing planning readiness with deployment authorization or operational success?

## Position in Architecture Governance Flow

## SLGA™ Governance Flow

## Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

## Cross-cutting: Software Lifecycle Control

## SDPS Core Module Flow

## DPM — Deployment Planning Module → DAM — Deployment Authorization Module

→ PVCM — Production Version Control Module → DSMM — Deployment State Monitoring Module → RRVM — Rollback & Recovery Module

DPM is the first Core Module within SDPS.

## It receives applicable release governance references from SRLS and

establishes the planning basis used by DAM to evaluate deployment authorization.

## Its planning references also support subsequent deployment execution,

production version identification, state monitoring, and recovery governance.

• Governed Space

## Software Deployment Planning Governance

## This space governs the formation, evaluation, maintenance, and reassessment

of plans for introducing identifiable software releases into defined operational environments.

It begins when a release is proposed for deployment into a particular environment.

It ends its initial planning cycle when the plan receives a supported classification and is submitted for authorization, deferred, rejected, or withdrawn from consideration.

Planning governance may reopen when relevant release, environment, authority, execution, operational, or recovery conditions change.

## Governed Object

The primary Governed Object is:

## A Software Deployment Plan representing the intended and

evidence-supported relationship between an identifiable software release, a defined operational environment, a deployment purpose, an execution strategy, applicable prerequisites, operational constraints, verification conditions, and intervention or recovery provisions.

The plan must preserve:

## Deployment Plan ID

## Deployment ID

## Software Product ID

## Release ID

## Target Environment ID

## Deployment Purpose

## Deployment Scope

## Deployment Strategy

## Execution Sequence

## Prerequisite References

## Resource Requirements

## Execution Actor Requirements

## Verification Conditions

## Intervention Conditions

## Rollback and Recovery References

## Planning State

Version

## Scope

DPM governs the methodological planning of software deployment.

It includes:

## Deployment Identification

## Purpose Definition

## Target Environment Definition

## Scope and Boundary Formation

## Strategy Selection

## Execution Sequence Planning

## Prerequisite Identification

## Operational Feasibility Assessment

## Resource and Capability Planning

## Actor and Tool Requirements

## Timing and Coordination

## Verification Planning

## Intervention Planning

## Rollback and Recovery Planning

## Planning Exceptions

## Plan Change Control

## Planning Evidence

## Plan Classification

## Authorization Handoff

DPM does not prescribe a particular deployment technology or organizational process.

# Foundational Principle

A deployment plan must establish what is intended, where it is intended, under which conditions it may be attempted, how its outcome will be verified, and how unacceptable outcomes will be addressed.

The existence of a technically executable deployment procedure does not establish that the deployment is sufficiently planned.

Deployment Script Available ≠ Deployment Plan Sufficient.

Deployment Plan Complete ≠ Deployment Authorized.

Deployment Technically Feasible ≠ Deployment Operationally Acceptable.

# Deployment Planning Identity

Every governed deployment plan must have a stable identity.

Required references include:

## Deployment Plan ID

## Deployment ID

## Software Product ID

## Release ID

## Target Environment ID

## Plan Version

## Plan Owner or Responsible Planning Function

## Planning Status

## Creation Time

## Last Material Revision

A plan may be revised without losing its historical identity.

Material revisions must remain distinguishable from earlier plan versions.

## Deployment Purpose Definition

Deployment purpose identifies the intended operational result.

Examples include:

## Initial Service Introduction

## Production Version Update

## Security Remediation Deployment

## Maintenance Release Deployment

## Emergency Correction

## Restricted Feature Introduction

## Staged Rollout

## Environment Migration

## Recovery Deployment

The purpose must be sufficiently specific to determine the relevant environment, execution requirements, verification criteria, and operational consequences.

A generic objective such as “deploy the latest version” may be insufficient where the release, target, or intended operational outcome remains ambiguous.

## Release Association

The plan must identify the software release proposed for deployment.

Relevant references include:

## Release ID

## Release Record Reference

## Release Authorization Reference

## Artifact Identity

## Release Restrictions

## Release Validity Conditions

## Known Release Limitations

## Deployment Eligibility Conditions

DPM must identify missing or conflicting release information.

It does not independently establish release readiness or grant release authorization.

## Target Environment Identification

The intended operational environment must be identifiable.

Relevant elements include:

## Environment ID

## Environment Type

## Infrastructure Reference

## Operational Purpose

## Environment Configuration

## Existing Production Version

## Relevant Dependencies

## Access Boundary

## Operational Constraints

## Environment State Reference

For distributed systems, the environment may consist of multiple regions, clusters, devices, or operational segments.

The plan must identify the deployment-relevant portions of that environment.

## Deployment Scope

Deployment scope defines the intended extent of the deployment.

It may include:

## Software Components

## Target Services

## Target Hosts or Devices

## Target Regions

## Target User Groups

## Affected Interfaces

## Required Configuration Changes

## Relevant Data Changes

## Operational Functions

## Expected Service Impact

The plan must distinguish included elements from excluded elements where that distinction materially affects execution or verification.

## Deployment Boundary

## The deployment boundary identifies what the plan permits for consideration

and what remains outside its intended scope.

Relevant boundaries include:

## Release Boundary

## Artifact Boundary

## Environment Boundary

## Infrastructure Boundary

## Configuration Boundary

## Dependency Boundary

## Actor Boundary

## Tool Boundary

## Temporal Boundary

## Operational Impact Boundary

## Recovery Boundary

DPM defines these boundaries for planning.

DAM determines whether the corresponding deployment actions are legitimately authorized.

## Existing Operational State

The plan must establish an adequate reference to the operational state that exists before deployment.

Relevant elements include:

## Current Production Version

## Current Artifact Identity

## Current Configuration

## Current Dependency State

## Current Service Health

## Current Infrastructure Conditions

## Current Operational Restrictions

## Known Incidents or Degradations

## Existing Recovery Conditions

## The required level of detail depends on deployment impact and operational

risk.

An unknown or unreliable baseline must be identified rather than treated as a verified starting condition.

## Intended Operational State

DPM must define the state the deployment is intended to establish.

Relevant elements include:

## Target Software Version

## Target Artifact Identity

## Target Configuration

## Target Dependency Conditions

## Target Activation State

## Target Service Availability

## Target Functional Conditions

## Target Operational Restrictions

## Target Verification Criteria

## The intended state provides the reference against which subsequent

deployment monitoring and verification can evaluate actual outcomes.

## State Transition Definition

The deployment plan must identify the intended transition between existing and target states.

## Existing Operational State → Planned Deployment Actions → Intended

Operational State

Relevant considerations include:

## Intermediate States

## Temporary Service Conditions

## Mixed-Version Operation

## Data or Schema Transitions

## Dependency Transitions

## Activation Sequence

## Traffic or Workload Movement

## Recovery Points

A deployment plan must not assume that intermediate states are harmless merely because the final intended state is acceptable.

## Deployment Strategy Selection

## DPM governs selection of a deployment strategy appropriate to the release

and environment.

Potential strategies include:

## Direct Replacement

## Rolling Deployment

## Canary Deployment

## Blue–Green Deployment

## Staged Deployment

## Parallel Deployment

## Feature-Gated Activation

## Immutable Infrastructure Replacement

## Fleet-Based Device Deployment

## Emergency Deployment

Strategy selection must consider the operational environment, deployment scope, risk, reversibility, resource availability, and verification capability.

## Strategy Suitability

Strategy suitability concerns whether the selected approach can reasonably establish the intended operational state under the identified conditions.

Relevant considerations include:

## Environment Compatibility

## Execution Feasibility

## Service Continuity

## Deployment Complexity

## Resource Requirements

## Observability

## Reversibility

## Failure Containment

## Recovery Feasibility

## Operational Restrictions

Strategy Technically Supported ≠ Strategy Suitable for the Deployment.

## Deployment Sequence

The plan must establish an execution sequence where order materially affects deployment outcomes.

Relevant elements include:

## Pre-Deployment Actions

## Environment Preparation

## Artifact Transfer

## Configuration Application

## Dependency Preparation

## Data or Schema Changes

## Software Activation

## Traffic Transition

## Verification Actions

## Completion Actions

## Recovery Checkpoints

Actions may be sequential, parallel, conditional, or event-triggered.

The plan must preserve the dependencies and conditions governing their execution order.

## Deployment Dependencies

DPM identifies dependencies required for successful execution of the plan.

These may include:

## Software Dependencies

## Infrastructure Dependencies

## Configuration Dependencies

## Service Dependencies

## Data Dependencies

## Network Dependencies

## External Provider Dependencies

## Operational Coordination Dependencies

## Tooling Dependencies

DPM governs their deployment-planning relevance.

The underlying software dependency governance remains within SDS.

## Deployment Prerequisites

Prerequisites define conditions that must be addressed before deployment authorization or execution.

Examples include:

## Applicable Release Authorization

## Artifact Availability

## Environment Availability

## Configuration Readiness

## Dependency Readiness

## Resource Availability

## Required Access

## Execution Capability

## Monitoring Capability

## Verification Capability

## Rollback or Recovery Preparedness

## Operational Coordination

The plan must identify which prerequisites are mandatory, conditional, or context-dependent.

## Prerequisite Condition Model

Each applicable prerequisite must retain an explicit condition state:

## Satisfied

## Not Satisfied

## Unknown

## Not Applicable

## Exception

A prerequisite must not be classified as satisfied solely because an automated system has not reported a failure.

Unknown ≠ Satisfied.

Exception ≠ Prerequisite Fulfilled.

Not Applicable ≠ Not Evaluated.

## Operational Feasibility

Operational feasibility evaluates whether the planned deployment can reasonably be performed within the actual conditions of the target environment.

Relevant dimensions include:

## Infrastructure Capacity

## Resource Availability

## Execution Capability

## Service Continuity Requirements

## Operational Constraints

## Dependency Conditions

## Maintenance Windows

## Recovery Capability

## Known Failure Conditions

## Required Coordination

Feasibility must remain evidence-supported and specific to the planned deployment.

## Resource Planning

DPM identifies resources required for deployment.

These may include:

## Compute Capacity

## Storage Capacity

## Network Capacity

## Deployment Tooling

## Required Credentials

## Human Operational Support

## Automation Capacity

## Monitoring Resources

## Backup Resources

## Recovery Resources

Resource availability must not be inferred solely from the theoretical capacity of the deployment technology.

## Execution Actor Requirements

The plan must identify the functions and capabilities required from execution actors.

Actors may include:

## Human Operators

## Deployment Teams

## Automated Pipelines

## Infrastructure Orchestrators

## Autonomous AI Agents

## External Service Providers

The plan should specify:

## Required Functions

## Required Competencies or Capabilities

## Required Access

## Tool Requirements

## Supervision Requirements

## Escalation Requirements

DPM identifies these requirements but does not independently grant execution authority.

## Tool and Execution Path Planning

Deployment plans involving automation or autonomous agents must identify material execution paths.

Relevant elements include:

## Deployment Tool

## Execution Interface

## Target Resource

## Required Permission

## Permitted Operation

## Data Access Requirement

## Configuration Access Requirement

## Downstream Action

## Execution Evidence Requirement

A deployment objective must not be treated as blanket permission for every tool path capable of achieving it.

Task Authority ≠ Constituent Action Authority.

## Autonomous Deployment Planning

Where autonomous agents participate in planning or execution, DPM must preserve the distinction between agent-generated proposals and legitimately approved deployment plans.

Relevant considerations include:

## Agent Identity

## Assigned Planning Function

## Planning Constraints

## Permitted Information Sources

## Proposed Execution Actions

## Unverified Assumptions

## Required Human or Institutional Review

## Escalation Conditions

## Planning Evidence

An agent may generate a technically coherent plan without establishing that its assumptions, authority references, or operational conditions are valid.

## Deployment Timing

The plan must identify relevant timing conditions.

These may include:

## Planned Start Time

## Expected Duration

## Maintenance Window

## Service Availability Constraints

## Dependency Timing

## Coordination Deadlines

## Verification Period

## Rollback Decision Window

## Recovery Time Constraints

Timing must reflect operational requirements rather than merely deployment pipeline convenience.

## Deployment Coordination

DPM governs the planning of coordination among affected actors and systems.

Relevant elements include:

## Affected Operational Functions

## Responsible Teams

## External Dependencies

## Required Notifications

## Execution Coordination

## Verification Responsibilities

## Escalation Contacts

## Recovery Coordination

## Communication Constraints

Coordination requirements must be proportionate to the deployment's operational scope and consequences.

## Service Continuity Planning

Where deployment may affect operational services, the plan must identify continuity requirements.

Relevant elements include:

## Permitted Downtime

## Service Availability Thresholds

## Traffic Management

## Redundancy Requirements

## Failover Conditions

## Temporary Degradation Limits

## Customer or User Impact

## Critical Operational Functions

## Recovery Requirements

A deployment strategy must not be considered acceptable solely because it can install the intended software.

## Data and State Transition Planning

Software deployment may require changes to persistent data or operational state.

DPM must identify applicable:

## Database Schema Changes

## Data Migration Requirements

## State Conversion

## Backward Compatibility Conditions

## Forward Compatibility Conditions

## Backup Requirements

## Data Integrity Verification

## Reversibility Limitations

## Recovery Constraints

The plan must distinguish software artifact rollback from restoration of compatible application and data states.

## Configuration Transition Planning

The plan must identify configuration changes required for the deployment.

Relevant elements include:

## Current Configuration Reference

## Target Configuration Reference

## Environment-Specific Parameters

## Configuration Dependencies

## Configuration Application Sequence

## Configuration Verification

## Configuration Recovery Conditions

## Known Configuration Restrictions

## DPM consumes configuration governance references from SCFS without

replacing that standard's responsibilities.

## Security Conditions

DPM must identify security-relevant conditions affecting deployment.

These may include:

## Artifact Integrity Requirements

## Deployment Credential Requirements

## Access Restrictions

## Environment Security Conditions

## Secret Management Requirements

## Network Exposure Changes

## Security Monitoring

## Known Vulnerability Restrictions

## Emergency Security Constraints

Security governance remains established through SSS and applicable security mechanisms.

DPM governs their relevance to the planned deployment.

## Deployment Risk Identification

The plan must identify material risks associated with deployment execution and its operational consequences.

Relevant categories include:

## Service Disruption Risk

## Artifact Mismatch Risk

## Configuration Failure Risk

## Dependency Failure Risk

## Data Incompatibility Risk

## Security Exposure Risk

## Partial Deployment Risk

## Monitoring Gap Risk

## Rollback Failure Risk

## Autonomous Execution Risk

Risk identification must be connected to applicable preventive, verification, intervention, or recovery provisions.

## Failure Scenario Planning

DPM must identify foreseeable deployment failure scenarios where their consequences materially affect planning.

Examples include:

## Artifact Transfer Failure

## Partial Installation

## Configuration Incompatibility

## Dependency Unavailability

## Activation Failure

## Database Migration Failure

## Service Health Degradation

## Unexpected Production Version

## Deployment Tool Failure

## Loss of Operational Connectivity

Failure Scenario Identified ≠ Failure Response Adequately Planned.

## Verification Planning

The deployment plan must specify how the resulting state will be evaluated.

Relevant elements include:

## Verification Objective

## Required Observations

## Artifact Identity Verification

## Production Version Verification

## Configuration Verification

## Dependency Verification

## Functional Verification

## Service Health Verification

## Evidence Sources

## Verification Responsibilities

## Verification Timing

## Acceptance Criteria

Verification must address the actual operational state rather than relying exclusively on execution completion signals.

## Monitoring Readiness

DPM must identify the monitoring capability required to observe deployment outcomes.

Relevant elements include:

## Monitoring Systems

## Required Metrics

## Required Logs

## Health Checks

## Version Observation

## Configuration Observation

## Failure Signals

## Alert Conditions

## Observation Coverage

## Monitoring Limitations

Monitoring Configured ≠ Monitoring Coverage Sufficient.

## DSMM governs the subsequent assessment of actual deployment-state

monitoring.

## Deployment Completion Criteria

The plan must define the conditions under which deployment execution may be considered complete.

Relevant criteria may include:

## Required Actions Executed

## Target Artifact Present

## Required Configuration Applied

## Required Services Activated

## Execution Exceptions Addressed

## Execution Evidence Preserved

## Verification Handoff Completed

Completion criteria must remain distinguishable from criteria for verified operational establishment.

## Operational Acceptance Criteria

## Operational acceptance criteria define the conditions required for the

resulting deployment state to be considered acceptable.

These may include:

## Verified Production Artifact

## Required Functional Behavior

## Service Availability

## Operational Health

## Configuration Consistency

## Dependency Compatibility

## Security Conditions

## Permitted Performance Range

## Residual Restrictions

## Evidence Sufficiency

DPM defines the intended criteria.

DSMM evaluates the actual operational state against them.

## Intervention Planning

DPM must identify conditions requiring intervention during or after deployment.

Potential triggers include:

## Critical Execution Failure

## Unauthorized Execution Action

## Unexpected Artifact Identity

## Operational Health Failure

## Unacceptable Service Degradation

## Material Configuration Deviation

## Security-Relevant Event

## Loss of Required Monitoring

## Recovery Threshold Exceeded

Planned responses may include pausing execution, restricting activation, stopping rollout, escalating, rolling back, or initiating recovery.

## Rollback Planning

Where rollback is relevant, the plan must identify:

## Rollback Objective

## Rollback Target

## Previous Artifact Availability

## Configuration Restoration Requirements

## Dependency Compatibility

## Data Compatibility

## Rollback Preconditions

## Rollback Execution Path

## Rollback Verification

## Rollback Limitations

A previous software version being available does not establish that rollback is feasible.

## Recovery Planning

DPM must identify recovery provisions for material deployment failure scenarios.

Recovery planning may include:

## Forward Correction

## Configuration Restoration

## Dependency Restoration

## Service Restart

## Infrastructure Recovery

## Data Restoration

## Traffic Redirection

## Alternative Operational Mode

## Operational Escalation

## Recovery Verification

## RRVM governs the subsequent rollback and recovery decisions and their

resulting states.

## Recovery Feasibility

Recovery feasibility must consider the actual constraints of the target environment.

Relevant dimensions include:

## Recovery Resource Availability

## Backup Availability

## Data Integrity

## Infrastructure Dependencies

## Software Compatibility

## Operational Time Constraints

## External Service Dependencies

## Required Expertise

## Recovery Verification Capability

## Known Irreversible Changes

Recovery Procedure Documented ≠ Recovery Feasible Under Actual Conditions.

## Irreversible Deployment Actions

Some deployment actions cannot be fully reversed.

Examples may include:

## Destructive Data Migration

## Irreversible Schema Transformation

## Firmware Changes

## External State Mutation

## Credential Revocation

## Permanent Data Deletion

## Physical Device State Changes

DPM must identify such actions and their operational consequences.

Where rollback is impossible or insufficient, the plan must establish appropriate forward-recovery or alternative response provisions.

## Deployment Checkpoints

## Deployment checkpoints define points at which execution or operational

conditions must be evaluated before proceeding.

Relevant elements include:

## Checkpoint ID

## Execution Stage

## Required Condition

## Evidence Source

## Decision Responsibility

## Permitted Continuation

## Failure Response

## Escalation Condition

Checkpoints are particularly important in staged, high-risk, autonomous, or partially irreversible deployments.

## Deployment Stop Conditions

The plan must identify conditions under which deployment execution must not begin or must not continue.

Examples include:

## Required Authorization Missing

## Mandatory Prerequisite Not Satisfied

## Artifact Identity Conflict

## Target Environment Mismatch

## Critical Security Restriction

## Insufficient Recovery Capability

## Unacceptable Operational Risk

## Loss of Required Verification Capability

## Execution Boundary Violation

A deployment stop condition must not be silently overridden by an automated system merely because technical execution remains possible.

## Planning Exceptions

DPM governs the identification and treatment of planning exceptions.

Relevant elements include:

## Exception ID

## Affected Planning Condition

## Reason

## Operational Consequence

## Proposed Treatment

## Authority Requirement

## Restriction

## Validity

## Reassessment Trigger

An exception recorded in the plan is not automatically an authorized exception.

## Exception approval belongs to the applicable decision and authority

mechanisms.

## Planning Uncertainty

DPM must explicitly represent material uncertainty.

Relevant examples include:

## Unknown Environment Condition

## Unverified Dependency

## Incomplete Resource Information

## Uncertain Deployment Duration

## Unknown Recovery Feasibility

## Unverified Monitoring Coverage

## Unresolved Data Compatibility

## Uncertain External Coordination

Unknown conditions must remain visible in the plan and influence its classification where applicable.

## Planning Evidence

Planning conclusions must be supported by relevant evidence.

Evidence may include:

## Release Records

## Environment Inventories

## Infrastructure Configuration Records

## Dependency References

## Capacity Assessments

## Deployment Test Results

## Monitoring Readiness Evidence

## Backup Verification

## Recovery Test Results

## Operational Constraints

## Technical Feasibility Assessments

Evidence must remain associated with the planning conclusion it supports.

## Evidence Sufficiency

DPM must evaluate whether the available evidence is sufficient to support the proposed deployment plan.

Relevant dimensions include:

## Release Identity Coverage

## Environment Coverage

## Prerequisite Coverage

## Execution Feasibility Evidence

## Resource Availability Evidence

## Verification Readiness Evidence

## Recovery Readiness Evidence

## Evidence Freshness

## Evidence Consistency

## Known Evidence Gaps

A detailed plan may still be insufficiently supported.

Plan Detail ≠ Evidence Sufficiency.

## Plan Consistency

DPM must evaluate consistency among the plan's principal elements.

Relevant relationships include:

## Release ↔ Deployment Purpose

## Release ↔ Target Environment

## Scope ↔ Deployment Strategy

## Strategy ↔ Operational Constraints

## Execution Sequence ↔ Dependencies

## Prerequisites ↔ Execution Requirements

## Monitoring ↔ Verification Criteria

## Failure Scenarios ↔ Recovery Provisions

## Actor Requirements ↔ Execution Paths

Material contradictions must be identified before the plan can receive a supported readiness classification.

## Plan Change Control

Material changes to the deployment plan must remain traceable.

Examples include:

## Release Change

## Target Environment Change

## Deployment Scope Change

## Strategy Change

## Execution Sequence Change

## Actor or Tool Change

## Operational Prerequisite Change

## Verification Condition Change

## Rollback or Recovery Change

## Timing Change

Each material change must preserve its reason, affected elements, evidence, and reassessment implications.

## Plan Version Continuity

DPM must preserve historical plan versions where their differences affect governance decisions.

Relevant elements include:

## Plan Version ID

## Previous Version Reference

## Changed Elements

## Change Reason

## Change Actor

## Change Time

## Affected Prerequisites

## Affected Evidence

## Reassessment Status

A revised plan must not silently replace the plan version associated with an earlier authorization decision.

## Planning Reassessment

Planning reassessment is required when material conditions change.

Triggers may include:

## Release Authorization Change

## Artifact Identity Change

### Environment Change

### Resource Availability Change

### Dependency Change

### Security Restriction Change

### Operational Constraint Change

### Monitoring Capability Change

### Recovery Feasibility Change

### New Material Risk Evidence

Reassessment must identify which planning conclusions remain applicable and which require revision.

## Planning Readiness Classification

DPM applies the following planning classifications:

### Not Initiated

### In Preparation

### Information Incomplete

### Prerequisites Unresolved

### Feasibility Not Established

### Evidence Insufficient

### Plan Inconsistent

### Planning Exception Pending

### Plan Ready for Authorization Assessment

### Plan Ready With Identified Conditions

### Plan Deferred

### Plan Rejected

### Plan Withdrawn

## Reassessment Required

## Superseded

## Conflicted

## Unknown

A plan classified as ready for authorization assessment is not thereby authorized for deployment.

## Governance Requirements

A conforming DPM implementation must satisfy the following requirements. Identity and Scope

• Establish a unique deployment plan identity

• Associate the plan with an identifiable deployment

• Identify the proposed software release

• Identify the target operational environment

• Define deployment purpose

• Define deployment scope and boundaries

• Preserve the existing operational state reference

• Define the intended operational state

## Strategy and Execution

• Establish a deployment strategy

• Evaluate strategy suitability

• Define the execution sequence

• Identify material execution dependencies

• Identify required actors and capabilities

• Identify material tool and execution paths

• Identify deployment timing constraints

• Identify coordination requirements

## Prerequisites and Feasibility

• Identify applicable deployment prerequisites

• Classify prerequisite condition states

• Evaluate operational feasibility

• Identify required resources

• Identify service continuity constraints

• Identify data and configuration transition requirements

• Identify security-relevant deployment conditions

• Preserve material uncertainties

## Verification and Recovery

• Define deployment verification conditions

• Identify required monitoring capabilities

• Define execution completion criteria

• Define operational acceptance criteria

• Identify intervention triggers

• Identify rollback requirements

• Evaluate rollback feasibility

• Identify recovery provisions

• Identify irreversible actions

• Define required execution checkpoints and stop conditions

## Evidence and Continuity

• Preserve planning evidence

• Evaluate evidence sufficiency

• Evaluate plan consistency

• Identify planning exceptions

• Preserve material plan revisions

• Maintain plan version continuity

• Identify reassessment triggers

• Establish planning readiness classification

• Preserve unresolved conditions

• Provide a traceable handoff to deployment authorization

## Governance Logic

## Identifiable Release → Deployment Purpose → Target Environment → Existing

State → Intended State → Deployment Scope → Deployment Strategy → Execution Sequence → Prerequisites → Feasibility → Actor and Tool Requirements → Verification Conditions → Intervention and Recovery Planning → Evidence Assessment → Plan Consistency → Planning Classification → Authorization Handoff

Where material conditions remain unresolved:

## Plan → Unresolved Condition → Reassessment / Revision / Deferral /

Rejection

Where material changes occur:

## Existing Plan Version → Material Change → Impact Identification → Evidence

Reassessment → Revised Plan Version → Updated Classification

DPM does not convert planning readiness into deployment authorization.

## Governance Outputs

The principal output is:

## Software Deployment Plan Profile

Supporting outputs may include:

## Deployment Plan Identity

## Deployment Purpose Record

## Target Environment Reference

## Deployment Scope and Boundary

## Existing Operational State Reference

## Intended Operational State Definition

## Deployment Strategy Record

## Execution Sequence

## Deployment Prerequisite Register

## Operational Feasibility Assessment

## Resource Requirement Record

## Actor and Tool Requirement Record

## Verification Plan

## Monitoring Readiness Reference

## Intervention Plan

## Rollback Plan Reference

## Recovery Plan Reference

## Planning Exception Register

## Planning Evidence Register

## Plan Consistency Assessment

## Plan Version History

## Planning Readiness Classification

## Authorization Handoff Record

## Minimum Implementation Framework

A minimum DPM implementation must establish eight governance layers. Layer 1 — Deployment Identity

## Identify the deployment plan, software release, deployment purpose, and

target environment. Layer 2 — Scope and State

## Define the deployment boundary, existing operational state, and intended

operational state. Layer 3 — Strategy and Execution

## Establish the deployment strategy, execution sequence, relevant actors,

tools, and dependencies. Layer 4 — Prerequisites and Feasibility

## Identify mandatory prerequisites, operational constraints, resource

requirements, and material feasibility conditions. Layer 5 — Verification

## Define execution completion criteria, operational acceptance conditions,

monitoring requirements, and evidence sources. Layer 6 — Intervention and Recovery

## Identify failure scenarios, stop conditions, rollback feasibility, recovery

provisions, and irreversible actions. Layer 7 — Evidence and Classification

## Preserve planning evidence, uncertainties, exceptions, consistency

findings, and planning readiness classification. Layer 8 — Continuity and Handoff

## Preserve plan versions, material revisions, reassessment conditions, and

the traceable handoff to DAM.

The implementation may be manual, software-supported, automated, or agent-assisted.

## The governance requirements remain applicable regardless of implementation

technology.

## Cross-Module Boundary

## DPM → DAM

DPM establishes the proposed deployment and its planning basis.

DAM evaluates whether the deployment is legitimately authorized.

## DPM → PVCM

DPM identifies the intended production version and target operational state.

## PVCM governs the identity and consistency of software versions actually

present in production.

## DPM → DSMM

## DPM defines verification conditions, monitoring requirements, and intended

operational outcomes.

## DSMM evaluates the actual deployment state and available operational

evidence.

## DPM → RRVM

DPM identifies intervention, rollback, and recovery provisions.

## RRVM governs the applicable rollback or recovery decisions, execution

references, and restoration verification.

Planning Ready ≠ Deployment Authorized.

Intended Production State ≠ Observed Production State.

Recovery Planned ≠ Recovery Successfully Executed.

## Cross-Architecture Boundary

DPM consumes relevant governance references from:

## SRLS — Software Release Standard

Release identity, authorization, records, and restrictions.

## SDS — Software Dependency Standard

Dependency identity, suitability, continuity, and risk references.

## SCFS — Software Configuration Standard

## Configuration baselines, build identities, and configuration change

references.

## SQS — Software Quality Standard

Quality conditions and relevant verification requirements.

## SSS — Software Security Standard

Security restrictions, vulnerabilities, and remediation-related conditions.

## GOA™ — Governance Architecture

Applicable authority and delegation references.

## ASGA™ — Autonomous Systems Governance Architecture

Autonomous execution boundaries and control references.

DPM does not redefine these mechanisms.

It governs their relevance to deployment planning.

## Practical Applications

## Enterprise Software Releases

Establishing structured deployment plans before production changes.

## Continuous Delivery

## Ensuring automated deployment procedures preserve prerequisites,

environment identity, verification conditions, and recovery planning.

## Autonomous AI Software Engineering

## Requiring AI-generated deployment plans to expose assumptions, execution

paths, authority requirements, and operational risks before authorization.

## Cloud-Native Systems

## Planning staged deployments, rolling updates, regional releases, and

multi-version transitions.

## Critical Infrastructure

## Establishing explicit operational constraints, stop conditions,

verification requirements, and recovery feasibility before deployment.

## Robotics and Embedded Systems

Planning software updates where irreversible firmware changes or altered physical behavior may affect recovery options.

## Use Case 1 — An AI Agent Produces a Technically Valid but Operationally

Incomplete Deployment Plan

## An autonomous software agent prepares a deployment plan for a production

application.

The plan includes executable commands and valid infrastructure credentials.

However, it does not identify the current production version, operational verification criteria, or rollback conditions.

The deployment is technically executable.

DPM evaluates:

## Release → Target Environment → Existing State → Execution Plan →

Verification Conditions → Recovery Provisions → Planning Evidence

The assessment identifies:

• Execution commands are available

• The target environment is identifiable

• The existing operational state is insufficiently established

• Operational verification criteria are incomplete

• Recovery feasibility is unknown

The appropriate classification is:

Information Incomplete — Plan Not Ready for Authorization Assessment.

Technical Executability ≠ Planning Sufficiency.

• Use Case 2 — A Deployment Plan Assumes Rollback Is Possible After an

Irreversible Database Migration

## A company prepares a software update requiring a database schema

transformation.

The plan includes a command to redeploy the previous application version.

However, the migration permanently changes stored data structures.

The previous version cannot operate correctly with the transformed database.

DPM evaluates:

## Current State → Data Transition → Intended State → Rollback Target →

Compatibility → Recovery Feasibility

The assessment identifies:

• The deployment sequence is defined

• The data transformation is materially irreversible

• Binary rollback does not restore data compatibility

• The stated rollback procedure is insufficient

• Alternative recovery provisions are required

The appropriate classification is:

Recovery Feasibility Not Established.

Previous Version Available ≠ Operational Rollback Feasible.

## Use Case 3 — A Global Deployment Plan Conceals Regional Operational

Differences

A software provider plans a simultaneous global rollout.

The plan identifies one release and one deployment procedure.

## However, regional environments have different dependency versions,

infrastructure capacity, and maintenance restrictions.

The plan assumes uniform conditions.

DPM evaluates:

## Release → Regional Environment Identity → Environment Conditions →

Strategy Suitability → Prerequisites → Execution Boundaries

The assessment identifies:

• The software release is identifiable

• The global deployment scope is defined

• Regional operating conditions materially differ

• The deployment strategy is not equally suitable across all

• Environment-specific planning conditions are required

environments.

The appropriate classification is:

Plan Reassessment Required — Environment-Specific Conditions.

One Release ≠ One Universally Suitable Deployment Plan.

## Critical Distinctions

Deployment Script Available ≠ Deployment Plan Sufficient.

Deployment Plan Complete ≠ Deployment Authorized.

Deployment Technically Feasible ≠ Deployment Operationally Acceptable.

Strategy Technically Supported ≠ Strategy Suitable for the Deployment.

Unknown ≠ Satisfied.

Exception ≠ Prerequisite Fulfilled.

Task Authority ≠ Constituent Action Authority.

Monitoring Configured ≠ Monitoring Coverage Sufficient.

Failure Scenario Identified ≠ Failure Response Adequately Planned.

Previous Version Available ≠ Rollback Feasible.

Recovery Procedure Documented ≠ Recovery Feasible Under Actual Conditions.

Plan Detail ≠ Evidence Sufficiency.

Planning Ready ≠ Deployment Authorized.

Intended Production State ≠ Observed Production State.

Recovery Planned ≠ Recovery Successfully Executed.

Technical Executability ≠ Planning Sufficiency.

One Release ≠ One Universally Suitable Deployment Plan.

## Canonical Boundary

## DPM governs the formation, evaluation, maintenance, and reassessment of

software deployment plans for identifiable releases entering defined operational environments.

## It establishes deployment purpose, target environment, scope, strategy,

execution sequence, prerequisites, feasibility, actor and tool requirements, verification conditions, intervention provisions, rollback and recovery planning, evidence, and planning readiness classification.

DPM does not independently authorize deployment, perform deployment execution, establish actual production version identity, verify operational deployment state, or execute rollback and recovery.

Its principal governance output is the:

## Software Deployment Plan Profile

## This output provides the planning basis for DAM and supporting references

for the remaining SDPS Core Modules.

## Canonical Closing Statement

A software deployment plan is not sufficient merely because its commands can execute.

A technically valid deployment procedure may target the wrong environment.

A complete execution sequence may depend on resources that are unavailable.

An automated pipeline may possess the necessary tools without possessing legitimate deployment authority.

A deployment may reach its intended software version while violating operational constraints.

And a rollback procedure may restore an earlier artifact without restoring a compatible operational state.

## DPM establishes the governance structure required to identify these

conditions before deployment authorization and execution.

## It requires the intended release, target environment, operational state

transition, deployment strategy, prerequisites, verification conditions, and recovery provisions to remain explicit and evidence-supported.

## It preserves planning uncertainty instead of converting assumptions into

verified facts.

## It requires material changes to trigger reassessment rather than silently

modifying the basis of a deployment decision.

And it ensures that human operators, automated pipelines, and autonomous agents remain subject to the same underlying planning requirements.

A deployment plan must demonstrate more than how software can be installed. It must establish what is intended to change, why the transition is appropriate, under which conditions it may proceed, how its outcome will be verified, and what must happen when the intended operational state cannot be achieved.

## DPM transforms deployment planning from a sequence of technical

instructions into a traceable, evidence-supported, and reassessable governance basis for legitimate software deployment.